

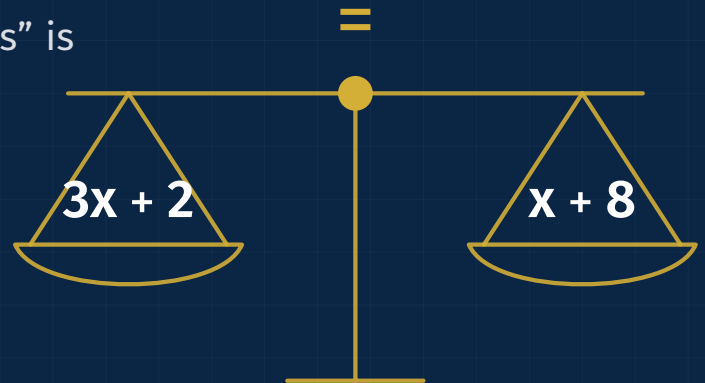


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CHAPTER 2 • ALGEBRA • TOPIC 11

Solving Inequalities

Solve them like equations — but flip the sign when you $\times$ or $\div$ by a negative.
Quadratics: “greater” is outside, “less” is between.



6

VARIANTS

20

PAST-PAPER PARTS

56

PRACTICE QUESTIONS

YOU WILL MASTER

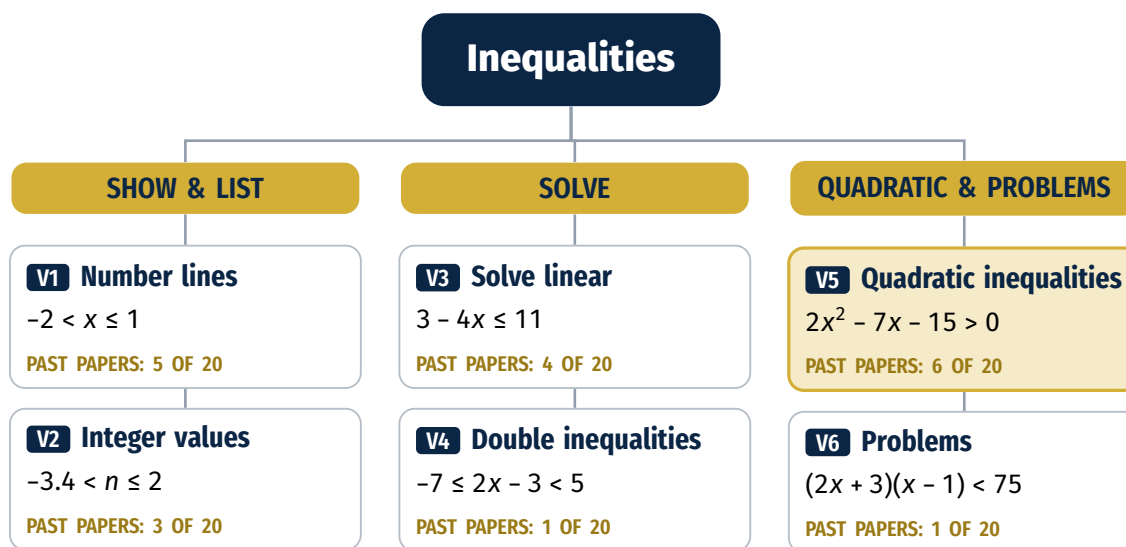
V1 Number lines V2 Integer values V3 Solve linear

V4 Double inequalities V5 Quadratic inequalities V6 Problems

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Inequalities at a glance



Your notes


THE ONE NEW RULE

× OR ÷ BY A NEGATIVE: FLIP

Everything else is solving an equation.

★ How often does each variant come up?

14 past-paper questions (Jan 2020 → Nov 2025) hold 20 inequality parts; 56 practice parts — grouped by type.

	PAST PAPERS (OUT OF 20)	PRACTICE PARTS
V1 Number lines	5	7
V2 Integer values	3	7
V3 Solve linear	4	27
V4 Double inequalities	1	2
V5 Quadratic inequalities	6	11
V6 Problems	1	2

WHAT THIS TELLS YOU Easy marks early in Paper 1 (V1–V3), and a 3-mark **quadratic inequality** late in Paper 2 (V5) — almost every recent series.

BEFORE YOU START

The big idea



An inequality has a range of answers.

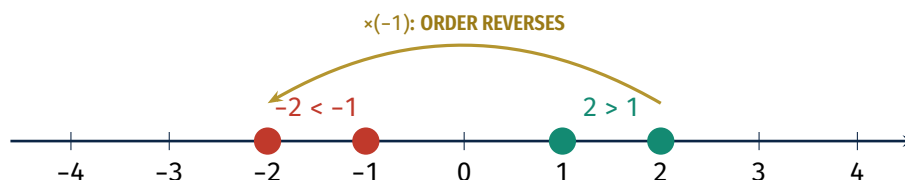
$x = 3$ is one number. $x > 3$ is **every** number bigger than 3 — a whole stretch of the number line.

Solve it **like an equation**. The only new rule:

multiply or divide by a negative → flip the sign.

sign	means	circle
$<$	less than	○ open
$>$	greater than	○ open
$\leq$	less than or equal	● filled
$\geq$	greater than or equal	● filled

Why the sign flips



CALCULATOR FIRST, THEN CHECK

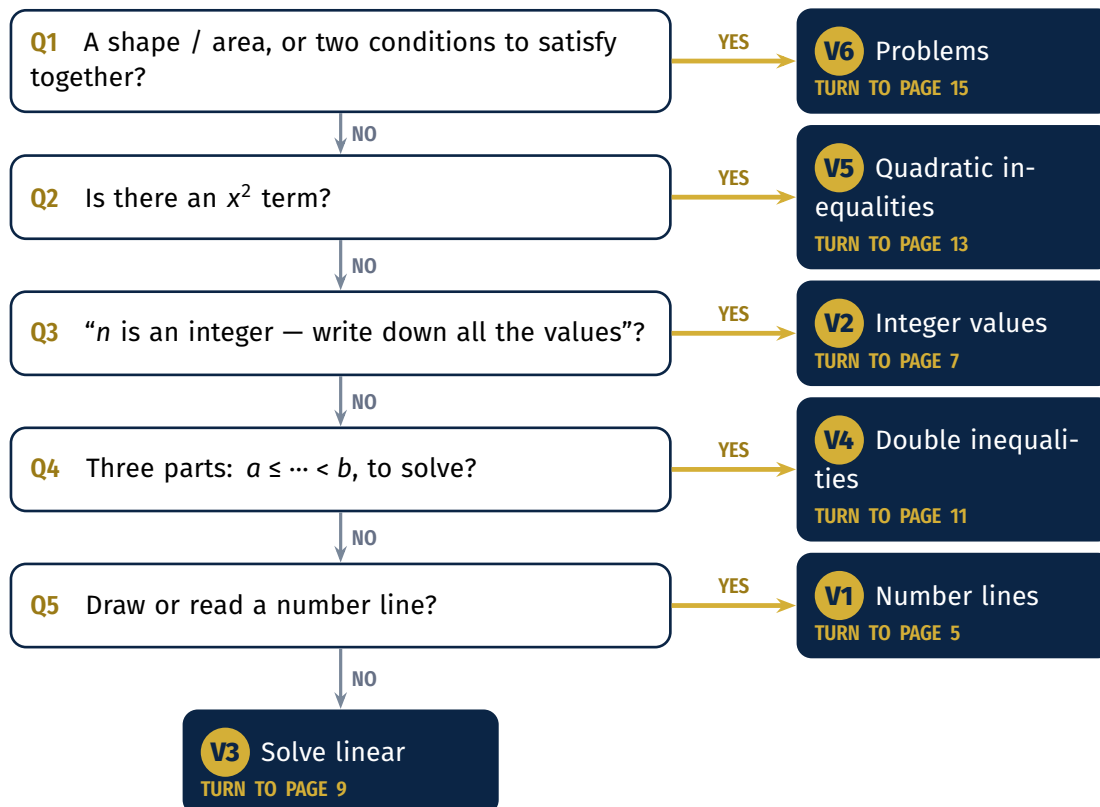
Check with a test value: pick a number inside your answer (e.g. $x = 0$ for $x > -2$) and put it into the question with **CALC**. True? Then try one outside — it must be false.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Number Lines

Show an inequality on a number line, or read one off it.

SEEN IN PAST PAPERS

5 / 20

PRACTICE QUESTIONS: 7

 Your notes

Came up in: Jan 2021 P1HR Q3(a) • Jan 2021 P2H Q5(c)(ii) • Nov 2021 P1H Q9(ii) • Jan 2022 P1HR Q1(b) • Jun 2025 P1HR Q6(a)

? What is it?

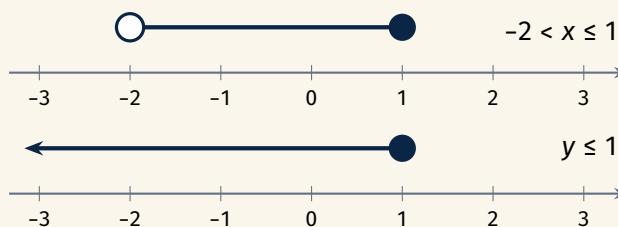
In plain words

A circle at each end, a line for the numbers in between.

Open circle ○: the number is **not** included ($<$, $>$).

Filled circle ●: it **is** included ($\leq$, $\geq$).

One end only? An arrow shows it goes on for ever.



IT LOOKS LIKE

$$-2 \leq y < 1$$

$$y \leq 1$$

$$-2 < x \leq 1$$

How do I do it?

1 Find the ends

The numbers next to the signs: -2 and 1.

2 Circle type

$< >$ → open ○. $\leq \geq$ → filled ●.

3 Join or arrow

Two ends: a line between. One end: an arrow the right way.

4 Reading one

Write the smaller number on the left: $-2 < x \leq 1$.

OPEN ○ $< >$ • **FILLED** ● $\leq \geq$ The line between the signs is "or equal".

V1 Number Lines

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

“On the number line, show the inequality”

→ **Circles (open/filled) and a line**

“Write down the inequality shown”

→ **Read both circles: $-2 < x \leq 1$**

“represent the solution set to part (i)”

→ **Solve first, then draw it**

WORKED EXAMPLE PAST PAPER

June 2025 · Paper 1HR · Q6(a) · 1 mark

Write down the inequality shown on the number line: an open circle at -2 , a filled circle at 1 , joined by a line.

STEP 1 the ends are -2 and 1

◀ find the ends

STEP 2 open at $-2 \rightarrow <$; filled at $1 \rightarrow \leq$

◀ circle type

STEP 4 $-2 < x \leq 1$

◀ smaller number on the left

ANSWER $-2 < x \leq 1$

✓ test $x = 0$: on the line ✓

WATCH OUT

× filled circle at -2 for $-2 < x$

✓ open circle at -2

$<$ does not include the end — the circle stays empty.

EXAM TIP

Draw the circles exactly above the numbers and use a ruler for the line — the mark is for both ends right.

Check yourself

calculator first · answers on page 17

1.1 Jan 2022 P1HR

On a number line, represent the inequality $y \leq 1$.

Answer:

1.2 Dr Eslam

Open circle at -3 , filled circle at 4 , line between. Write the inequality.

Answer:

1.3 Nov 2021 P1H

Represent $-2 \leq x < 4$ on a number line.

Answer:

V2

VARIANT 2

Integer Values

“ n is an integer”: list every whole number in the range — check both ends.

SEEN IN PAST PAPERS

3 / 20

PRACTICE QUESTIONS: 7

 Your notes

Came up in: Jan 2021 P1HR Q3(b) • May 2021 P1H Q7(a) • Jan 2022 P1HR Q1(a)

? What is it?

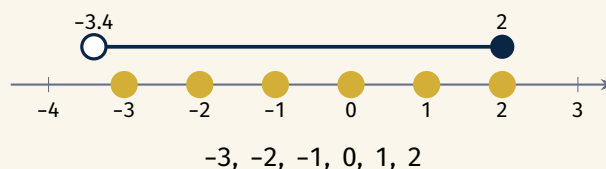
In plain words

Integers are whole numbers:

..., -2, -1, 0, 1, 2, ... List **every** one in the range.

Look at each end: $\leq$ **includes it**, $<$ does not. Do not forget **0** and the negatives.

If there is a number in front ($-4 \leq 2y < 6$), solve first.



IT LOOKS LIKE

$$-3.4 < n \leq 2$$

$$-2 \leq n < 3$$

$$-4 \leq 2y < 6$$

How do I do it?

1 Solve if needed

$$-4 \leq 2y < 6 \rightarrow \div 2 \text{ all three: } -2 \leq y < 3.$$

2 Check the ends

$\leq \rightarrow$ include the end; $< \rightarrow$ leave it out.

3 List them all

Count up from the bottom, through 0, to the top.

SAY EVERY INTEGER OUT LOUD ...including 0.

V2 Integer Values

VARIANT 2 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO*"n is an integer. Write down all the values of n"*→ **List them; check both ends***" $-4 \leq 2y < 6$, y is an integer"*→ **Solve first:** $-2 \leq y < 3$ *"the largest integer that satisfies"*→ **Solve, then the integer just inside****WORKED EXAMPLE** **PAST PAPER**

January 2021 · Paper 1HR · Q3(b)

*n is an integer. Write down all the values of n that satisfy $-3.4 < n \leq 2$.***STEP 2** bottom: -3.4 , not an integer → start at -3 ◀ check the ends**STEP 2** top: $\leq 2 \rightarrow 2$ is included ◀ check the ends**STEP 3** $-3, -2, -1, 0, 1, 2$ ◀ list all, through 0**ANSWER** $-3, -2, -1, 0, 1, 2$

✓ six integers

WATCH OUT✗ $-3, -2, -1, 1, 2$ ✓ $-3, -2, -1, 0, 1, 2$

0 is an integer — the most forgotten one.

EXAM TIPCount them: from -3 to 2 is six numbers. If you have five, one is missing.**Check yourself**

calculator first · answers on page 17

2.1 Jan 2022 P1HR*n is an integer. Write down all n such that $-2 \leq n < 3$.*

Answer:

2.2 May 2021 P1H *$-4 \leq 2y < 6$, y is an integer. Write down all the possible values of y.*

Answer:

2.3 Practice · Easy*Find the largest integer x with $x < 2.2$.*

Answer:

V3

VARIANT 3

Solve a Linear Inequality

Clean, then isolate — exactly like an equation. Flip the sign only for $\times$ or $\div$ by a negative.

SEEN IN PAST PAPERS

4 / 20

PRACTICE QUESTIONS: 27



Your notes

Came up in: Jan 2021 P2H Q5(c)(i) • May 2021 P1H Q7(b) • Jun 2022 P1HR Q9 • Jun 2025 P1HR Q6(b)

? What is it?

In plain words

Use Dr Eslam's equation method (Topic 9): **Clean** → **Isolate**. The inequality sign stays as it is —

unless you multiply or divide by a negative: then it **flips**.

Tip: move the x terms so the x stays positive — then you never divide by a negative.

$$-4x \leq 8$$

↓ $\div(-4)$: **FLIP** $\leq$ **TO** $\geq$

$$x \geq -2$$

IT LOOKS LIKE

$$7t - 8 < 2t + 7$$

$$3 - 4x \leq 11$$

$$7a - 5 \leq 3a + 28$$

How do I do it?

1 Clean

Expand brackets, remove fractions ($\times$ the LCM — positive, no flip).

2 Collect

x terms one side (keep x positive if you can), numbers the other.

3 Divide

Divide by the number in front of x .
Negative? Flip the sign.

4 Check

Test a value from your answer in the question.

$\times$ OR $\div$ BY A NEGATIVE: FLIP Adding or subtracting never flips.

V3 Solve a Linear Inequality

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Solve the inequality"

→ Same steps as an equation

Dividing by -4 → Flip: $\leq$ becomes $\geq$

"Is her solution correct?"

→ Solve it yourself; compare sign and circle

**WORKED EXAMPLE** **PAST PAPER**

June 2022 • Paper 1HR • Q9 • 2 marks

Solve the inequality $3 - 4x \leq 11$ **STEP 2** $-4x \leq 11 - 3$

◀ take 3 off both sides

STEP 2 $-4x \leq 8$

◀ simplify

STEP 3 $x \geq \frac{8}{-4} = -2$ ◀ divide by -4 : **flip****ANSWER** $x \geq -2$ ✓ test $x = 0$: $3 \leq 11$ ✓**WATCH OUT**✗ $-4x \leq 8 \Rightarrow x \leq -2$ ✓ $-4x \leq 8 \Rightarrow x \geq -2$ Dividing by a negative reverses the order
— the sign must flip.**EXAM TIP**Avoid the flip: $3 - 4x \leq 11 \rightarrow 3 - 11 \leq 4x$
 $\rightarrow -8 \leq 4x \rightarrow -2 \leq x$. Same answer, no
negative divide.**Check yourself**

calculator first • answers on page 17

3.1

Jan 2021 P2H

Solve $7t - 8 < 2t + 7$

Answer:

3.2

Jun 2025 P1HR

Solve $7a - 5 \leq 3a + 28$

Answer:

3.3

May 2021 P1H

Solve $7t - 3 \leq 2t + 31$

Answer:

V4

VARIANT 4

Double Inequalities

Three parts: whatever you do, do it to all three.

SEEN IN PAST PAPERS

1 / 20

PRACTICE QUESTIONS: 2



Your notes

Came up in: Nov 2021 P1H Q9(i)

? What is it?**In plain words**

$-7 \leq 2x - 3 < 5$ has **three parts**. Get x alone in the **middle** by doing the same thing to **all three** parts.

The answer is a range: $-2 \leq x < 4$ — often then drawn on a number line or listed as integers.

IT LOOKS LIKE

$$-7 \leq 2x - 3 < 5$$

$$-5 < 3x + 1 \leq 10$$

$$\boxed{-7} \leq \boxed{2x - 3} < \boxed{5}$$

+3 TO ALL THREE

$$\boxed{-4} \leq \boxed{2x} < \boxed{8}$$

÷2 TO ALL THREE

$$\boxed{-2} \leq \boxed{x} < \boxed{4}$$

How do I do it?**1 Add / subtract**Undo the number with x — in all three parts.**2 Divide**Divide all three parts by the number in front of x .**3 Read the range** $-2 \leq x < 4$: keep each sign where it was.**DO IT TO ALL THREE** Then draw it or list the integers.

V4 Double Inequalities

VARIANT 4 • SEE IT DONE, THEN TRY


 Your notes

WHEN YOU SEE → YOU DO

"Solve the inequalities $-7 \leq 2x - 3 < 5$ " → **All three parts, x alone in the middle**

"... represent the solution set" → **Filled at -2 , open at 4**

"... n is an integer" → **Solve, then list (V2)**

WORKED EXAMPLE PAST PAPER

November 2021 • Paper 1H • Q9(i)

Solve the inequalities $-7 \leq 2x - 3 < 5$

STEP 1 $-7 + 3 \leq 2x < 5 + 3$

◀ add 3 to all three parts

STEP 1 $-4 \leq 2x < 8$

◀ simplify

STEP 2 $-2 \leq x < 4$

◀ divide all three by 2

ANSWER $-2 \leq x < 4$

✓ test $x = 0$: $-7 \leq -3 < 5$ ✓

WATCH OUT

✗ $-7 \leq 2x < 5 + 3$

✓ $-7 + 3 \leq 2x < 5 + 3$

Every part gets the same operation — not just the right side.

EXAM TIP

Show the line $-4 \leq 2x < 8$ before dividing — that line is the method mark.

Check yourself

calculator first • answers on page 17

4.1

Dr Eslam

Solve $-5 < 3x + 1 \leq 10$

Answer:

4.2

Dr Eslam

$-3 \leq 2n + 1 < 7$, n an integer. List the values of n .

Answer:

4.3

Dr Eslam

Solve $1 < \frac{x+5}{2} \leq 4$

Answer:

V5**VARIANT 5** MOST COMMON IN THE EXAM

Quadratic Inequalities

Factorise, sketch the $\cup$: “greater than 0” is outside the roots, “less than 0” is between.

SEEN IN PAST PAPERS

6 / 20

PRACTICE QUESTIONS: 11



Your notes

Came up in: Jan 2020 P2HR Q19(b) • Jun 2024 P2HR Q20 • Nov 2024 P2H Q20 • Jun 2025 P2HR Q21 • Jun 2025 P2H Q19 • Nov 2025 P1H Q25

? What is it?

In plain words

Find the **roots** (where it = 0), sketch the $\cup$ -curve through them, and read the answer:

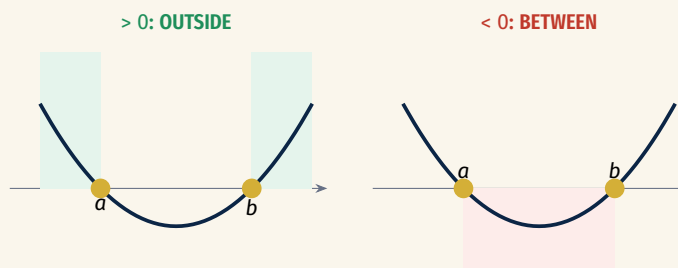
■ > 0 : the curve is **above** the axis $\rightarrow$ **outside**: $x < a$ or $x > b$.

■ < 0 : **below** the axis $\rightarrow$ **between**: $a < x < b$.

IT LOOKS LIKE

$$2x^2 - 7x - 15 > 0$$

$$4x^2 + 4x - 15 < 0$$



How do I do it?

1 ERCSMake it $\dots > 0$ or $\dots < 0$ with a positive x^2 .**2** RootsCalculator, then factorise: $(2x + 3)(x - 5) \rightarrow$ roots -1.5 and 5 .**3** SketchA $\cup$ through the roots. Shade above (> 0) or below (< 0).**4** Write it > 0 : $x < a$ or $x > b$. < 0 : $a < x < b$.**GREATER: OUTSIDE • LESS: BETWEEN**

Dr Eslam's quick read.

V5 Quadratic Inequalities

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO $\dots > 0$ ("greater than")→ **Outside:** $x < a$ or $x > b$ $\dots < 0$ ("less than")→ **Between:** $a < x < b$

"Show clear algebraic working"

→ **Factorised form + a sketch with the roots****WORKED EXAMPLE** **PAST PAPER**

June 2025 · Paper 2HR · Q21 · 3 marks

Solve the inequality $2x^2 - 7x - 15 > 0$. Show clear algebraic working.**STEP 2** $(2x + 3)(x - 5) > 0$ ◀ calculator roots $-1.5, 5 \rightarrow$
brackets**STEP 3** sketch a $\cup$ through -1.5 and 5 ; we want
above the axis◀ > 0 : outside**STEP 4** $x < -1.5$ or $x > 5$

◀ write the two regions

ANSWER $x < -1.5$ or $x > 5$ ✓ test $x = 0$: $-15 > 0$ false ✓**WATCH OUT**✗ $x^2 > 9 \Rightarrow x > 3$ ✓ $x^2 > 9 \Rightarrow x < -3$ or $x > 3$ Never square-root an inequality —
factorise and sketch.**EXAM TIP**Test $x = 0$ in the question: if it is true, 0
is in your answer; if false, it is not. A
quick check of "inside or outside".**Check yourself**

calculator first · answers on page 17

5.1

Jun 2025 P2H

Solve $4x^2 + 4x - 15 < 0$

Answer:

5.2

Nov 2024 P2H

Solve
 $10x^2 + 11x - 21 < 0$

Answer:

5.3

Jun 2024 P2HR

Solve $6x^2 - 7x - 20 > 0$

Answer:

V6

VARIANT 6

Problems with Inequalities

A shape or two conditions: build the inequality, solve, then keep the overlap.

SEEN IN PAST PAPERS

1 / 20

PRACTICE QUESTIONS: 2

 Your notes

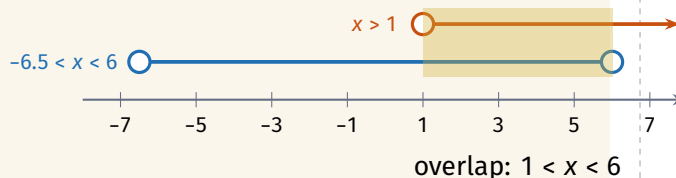
Came up in: Jan 2022 P2H Q22

? What is it?

In plain words

"The area is **less than 75**" → area formula < 75 , then V5.

A length must be **positive**: $(x - 1) > 0$ is a second condition. Draw both on one number line and keep the **overlap**.



IT LOOKS LIKE

$$(2x + 3)(x - 1) < 75$$

$$n \leq 4 \text{ and } 1 < n < 5$$

How do I do it?

1 Build it

Area formula < 75 (or the words) as an inequality.

2 Solve

ERCS, factorise, sketch (V5).

3 Other conditions

Lengths > 0 , integers, the question's extra rule.

4 Overlap

Draw them all on one number line; keep what satisfies all.

KEEP THE OVERLAP Every condition must be true.

V6 Problems with Inequalities

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"the area is less than 75 cm^2 "*→ **Area < 75 , then solve***A side like $(x - 1) \text{ cm}$* → **It must be positive: $x > 1$** *"integer such that ... and ..."*→ **Solve both, list the integers in the overlap****WORKED EXAMPLE** **PAST PAPER**

January 2022 · Paper 2H · Q22 · 5 marks

A rectangle is $(2x + 3) \text{ cm}$ by $(x - 1) \text{ cm}$. Its area is less than 75 cm^2 . Find the range of possible values of x .

STEP 1 $(2x + 3)(x - 1) < 75 \Rightarrow 2x^2 + x - 78 < 0$ ◀ build, ERCS**STEP 2** $(2x + 13)(x - 6) < 0 \Rightarrow -6.5 < x < 6$ ◀ roots $-6.5, 6$; < 0 : between**STEP 3** $x - 1 > 0 \Rightarrow x > 1$ ◀ a length is positive**STEP 4** $1 < x < 6$ ◀ keep the overlap**ANSWER** $1 < x < 6$ ✓ $x = 2: 7 \times 1 = 7 < 75$ ✓**WATCH OUT**✗ answer $-6.5 < x < 6$ ✓ $1 < x < 6$

The side $(x - 1)$ would be negative for $x < 1$ — the shape needs it positive.

EXAM TIP

5 marks: inequality, factorised, the range, the length condition, the final overlap. Write each.

Check yourself

calculator first · answers on page 17

6.1

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n is an integer with $2n + 1 > 3$ and $3n - 2 \leq 10$. List the values of n .

Answer:

6.2

Practice · Hard

Solve $6x^2 - 11x - 35 < 0$ given that $x > 2$.

Answer:

6.3

Dr Eslam

A square has side $(x + 2) \text{ cm}$ and area less than 49 cm^2 . Find the range of x .

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 filled circle at 1,
arrow to the left

1.2 $-3 < x \leq 4$

1.3 filled at -2, open at
4, line between

2.1 -2, -1, 0, 1, 2

2.2 -2, -1, 0, 1, 2

2.3 2

3.1 $t < 3$

3.2 $a \leq 8.25$

3.3 $t \leq 6.8$

4.1 $-2 < x \leq 3$

4.2 -2, -1, 0, 1, 2

4.3 $-3 < x \leq 3$

5.1 $-2.5 < x < 1.5$

5.2 $-2.1 < x < 1$

5.3 $x < -\frac{4}{3}$ or $x > \frac{5}{2}$

6.1 2, 3, 4

6.2 $2 < x < 3.5$

6.3 $-2 < x < 5$

THE WHOLE TOPIC ON ONE PAGE

Inequalities cheat sheet



Your notes
V1 Number lines 5/20

$$-2 < x \leq 1$$

< > open ○, $\leq \geq$ filled ●.**V2 Integer values 3/20**

$$-3.4 < n \leq 2 : -3, \dots, 2$$

Check both ends; include 0.

V3 Solve linear 4/20

$$-4x \leq 8 \Rightarrow x \geq -2$$

Like an equation; $\times \div$ negative $\rightarrow$ flip.**V4 Double 1/20**

$$-7 \leq 2x - 3 < 5 \Rightarrow -2 \leq x < 4$$

Do it to all three parts.

V5 Quadratic 6/20

$$(2x + 3)(x - 5) > 0$$

Greater: outside. Less: between. Sketch.

V6 Problems 1/20

$$-6.5 < x < 6, x > 1 \Rightarrow 1 < x < 6$$

Build, solve, keep the overlap.

**TOP 3 LOST MARKS**

1. not flipping for $\div$ negative $\times$
2. forgetting 0 in an integer list $\times$
3. $x^2 > 9 \Rightarrow x > 3$ only $\times$

**CALCULATOR FIRST, THEN CHECK**

Roots from equation mode. Then test a value inside your answer (true) and one outside (false).

BOTTOM LINE Solve like an equation • $\times \div$ negative: flip • quadratic: greater outside, less between • keep the overlap.